B The multivariate Gaussian family: An exponential family

We report the canonical decomposition of the multivariate Gaussian [136] family $\{N(\mu, \Sigma)$ such that $\mu \in \mathbb{R}^d, \Sigma \succ 0\}$ following [78]. The multivariate Gaussian family is also called the *MultiVariate Normal* family, or MVN family for short.

Let $\lambda := (\lambda_v, \lambda_M) = (\mu, \Sigma)$ denote the *composite* (vector,matrix) parameter of an MVN. The d-dimensional MVN density is given by

$$p_\lambda(x; \lambda) \quad := \quad \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{|\lambda_M|}} \exp\left(-\frac{1}{2}(x-\lambda_v)^\top \lambda_M^{-1}(x-\lambda_v)\right), \tag{221}$$

where $|\cdot|$ denotes the matrix determinant. The natural parameters θ are also expressed using both a vector parameter θ_v and a matrix parameter θ_M in a compound object $\theta = (\theta_v, \theta_M)$. By defining the following *compound inner product* on a composite (vector,matrix) object

$$\langle \theta, \theta' \rangle := \theta_v^\top \theta'_v + \mathrm{tr}\left({\theta'_M}^\top \theta_M\right), \tag{222}$$

where $\mathrm{tr}(\cdot)$ denotes the matrix trace, we rewrite the MVN density of Equation (221) in the canonical form of an exponential family [84]:

$$p_\theta(x; \theta) \quad := \quad \exp\left(\langle t(x), \theta\rangle - F_\theta(\theta)\right) = p_\lambda(x; \lambda(\theta)), \tag{223}$$

where

$$\theta = (\theta_v, \theta_M) = \left(\Sigma^{-1}\mu, -\frac{1}{2}\Sigma^{-1}\right) = \theta(\lambda) = \left(\lambda_M^{-1}\lambda_v, -\frac{1}{2}\lambda_M^{-1}\right), \tag{224}$$

is the *compound natural parameter* and

$$t(x) = (x, -xx^\top) \tag{225}$$

is the *compound sufficient statistic*. The function F_θ is the strictly convex and continuously differentiable log-normalizer defined by:

$$F_\theta(\theta) = \frac{1}{2}\left(d\log\pi - \log|\theta_M| + \frac{1}{2}\theta_v^\top \theta_M^{-1}\theta_v\right), \tag{226}$$

The log-normalizer can be expressed using the ordinary parameters, $\lambda = (\mu, \Sigma)$, as:

$$F_\lambda(\lambda) \quad = \quad \frac{1}{2}\left(\lambda_v^\top \lambda_M^{-1}\lambda_v + \log|\lambda_M| + d\log 2\pi\right), \tag{227}$$

$$= \quad \frac{1}{2}\left(\mu^\top \Sigma^{-1}\mu + \log|\Sigma| + d\log 2\pi\right). \tag{228}$$

The *moment/expectation parameters* [8] are

$$\eta = (\eta_v, \eta_M) = E[t(x)] = \nabla F(\theta). \tag{229}$$

We report the conversion formula between the three types of coordinate systems (namely the ordinary parameter λ, the natural parameter θ and the moment parameter η) as follows:

$$\left\{\begin{array}{l} \theta_v(\lambda) = \lambda_M^{-1}\lambda_v = \Sigma^{-1}\mu \\ \theta_M(\lambda) = \frac{1}{2}\lambda_M^{-1} = \frac{1}{2}\Sigma^{-1} \end{array}\right. \quad \Leftrightarrow \quad \left\{\begin{array}{l} \lambda_v(\theta) = \frac{1}{2}\theta_M^{-1}\theta_v = \mu \\ \lambda_M(\theta) = \frac{1}{2}\theta_M^{-1} = \Sigma \end{array}\right. \tag{230}$$

$$\left\{\begin{array}{l} \eta_v(\theta) = \frac{1}{2}\theta_M^{-1}\theta_v \\ \eta_M(\theta) = -\frac{1}{2}\theta_M^{-1} - \frac{1}{4}(\theta_M^{-1}\theta_v)(\theta_M^{-1}\theta_v)^\top \end{array}\right. \quad \Leftrightarrow \quad \left\{\begin{array}{l} \theta_v(\eta) = -(\eta_M + \eta_v\eta_v^\top)^{-1}\eta_v \\ \theta_M(\eta) = -\frac{1}{2}(\eta_M + \eta_v\eta_v^\top)^{-1} \end{array}\right. \tag{231}$$

$$\left\{\begin{array}{l} \lambda_v(\eta) = \eta_v = \mu \\ \lambda_M(\eta) = -\eta_M - \eta_v\eta_v^\top = \Sigma \end{array}\right. \quad \Leftrightarrow \quad \left\{\begin{array}{l} \eta_v(\lambda) = \lambda_v = \mu \\ \eta_M(\lambda) = -\lambda_M - \lambda_v\lambda_v^\top = -\Sigma - \mu\mu^\top \end{array}\right. \tag{232}$$

The dual Legendre convex conjugate [8] is

$$F_\eta^*(\eta) = -\frac{1}{2}\left(\log(1+\eta_v^\top \eta_M^{-1}\eta_v) + \log|-\eta_M| + d(1+\log 2\pi)\right), \tag{233}$$

and $\theta = \nabla_\eta F_\eta^*(\eta)$.

We check the Fenchel-Young equality when $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$:

$$F_\theta(\theta) + F_\eta^*(\eta) - \langle \theta, \eta\rangle = 0. \tag{234}$$

The KullbackLeibler divergence between two d-dimensional Gaussians distributions $p_{(\mu_1,\Sigma_1)}$ and $p_{(\mu_2,\Sigma_2)}$ (with $\Delta_\mu = \mu_2 - \mu_1$) is

$$\mathrm{KL}(p_{(\mu_1,\Sigma_1)} : p_{(\mu_2,\Sigma_2)}) \quad = \quad \frac{1}{2}\left\{\mathrm{tr}(\Sigma_2^{-1}\Sigma_1) + \Delta_\mu^\top \Sigma_2^{-1}\Delta_\mu + \log\frac{|\Sigma_2|}{|\Sigma_1|} - d\right\} = \mathrm{KL}(p_{\lambda_1} : p_{\lambda_2}). \tag{235}$$

We check that $\mathrm{KL}(p_{(\mu,\Sigma)} : p_{(\mu,\Sigma)}) = 0$ since $\Delta_\mu = 0$ and $\mathrm{tr}(\Sigma^{-1}\Sigma) = \mathrm{tr}(I) = d$. Notice that when $\Sigma_1 = \Sigma_2 = \Sigma$, we have

$$\mathrm{KL}(p_{(\mu_1,\Sigma)} : p_{(\mu_2,\Sigma)}) = \frac{1}{2}\Delta_\mu^\top \Sigma^{-1}\Delta_\mu = \frac{1}{2}D^2_{\Sigma^{-1}}(\mu_1,\mu_2), \tag{236}$$

that is half the squared Mahalanobis distance for the precision matrix Σ^{-1} (a positive-definite matrix: $\Sigma^{-1} \succ 0$), where the Mahalanobis distance is defined for any positive matrix $Q \succ 0$ as follows:

$$D_Q(p_1 : p_2) = \sqrt{(p_1 - p_2)^\top Q(p_1 - p_2)}. \tag{237}$$

The KullbackLeibler divergence between two probability densities of the same exponential families amount to a Bregman divergence [8]:

$$\mathrm{KL}(p_{(\mu_1,\Sigma_1)} : p_{(\mu_2,\Sigma_2)}) = \mathrm{KL}(p_{\lambda_1} : p_{\lambda_2}) = B_F(\theta_2 : \theta_1) = B_{F^*}(\eta_1 : \eta_2), \tag{238}$$

where the Bregman divergence is defined by

$$B_F(\theta : \theta') := F(\theta) - F(\theta') - \langle \theta - \theta', \nabla F(\theta')\rangle, \tag{239}$$

with $\eta' = \nabla F(\theta')$. Define the canonical divergence [8]

$$A_F(\theta_1 : \eta_2) = F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2\rangle = A_{F^*}(\eta_2 : \theta_1), \tag{240}$$

since $F^{**} = F$. We have $B_F(\theta_1 : \theta_2) = A_F(\theta_1 : \eta_2)$.

Notations

Below is a list of notations we used in this document:

$[D]$	$[D] := \{1,\ldots,D\}$
$\langle\cdot,\cdot\rangle$	inner product
$M_Q(u,v) = \|u-v\|_Q$	Mahalanobis distance $M_Q(u,v) = \sqrt{\sum_{i,j}(u^i - v^i)(u^j - v^j)Q_{ij}}$, $Q \succ 0$
$D(\theta : \theta')$	parameter divergence
$D[p(x) : p'(x)]$	statistical divergence
D, D^*	Divergence and dual (reverse) divergence

Csiszár divergence I_f	$I_f(\theta:\theta') := \sum_{i=1}^D \theta_i f\left(\frac{\theta'_i}{\theta_i}\right)$ with $f(1)=0$
Bregman divergence B_F	$B_F(\theta:\theta') := F(\theta) - F(\theta') - (\theta-\theta')^\top \nabla F(\theta')$
Canonical divergence A_{F,F^*}	$A_{F,F^*}(\theta:\eta') = F(\theta) + F^*(\eta') - \theta^\top \eta'$
Bhattacharyya distance	$B_\alpha[p_1:p_2] = -\log \int_{x\in\mathcal{X}} p_1^\alpha(x) p_2^{1-\alpha}(x) \mathrm{d}\mu(x)$
Jensen/Burbea-Rao divergence	$J_F^{(\alpha)}(\theta_1:\theta_2) = \alpha F(\theta_1) + (1-\alpha)F(\theta_2) - F(\theta_1 + (1-\alpha)\theta_2)$
Chernoff information	$C[P_1,P_2] = -\log \min_{\alpha\in(0,1)} \int_{x\in\mathcal{X}} p_1^\alpha(x) p_2^{1-\alpha}(x) \mathrm{d}\mu(x)$
F, F^*	Potential functions related by Legendre-Fenchel transformation
$D_\rho(p,q)$	Riemannian distance $D_\rho(p,q) := \int_0^1 \lVert\gamma'(t)\rVert_{\gamma(t)} \mathrm{d}t$
B, B^*	basis, reciprocal basis
$B = \{e_1 = \partial_1, \ldots, e_D = \partial_D\}$	natural basis
$\{\mathrm{d}x_i\}_i$	covector basis (one-forms)
$(v)_B := (v^i)$	contravariant components of vector v
$(v)_{B^*} := (v_i)$	covariant components of vector v
$u \perp v$	vector u is perpendicular to vector v ($\langle u,v\rangle = 0$)
$\lVert v\rVert = \sqrt{\langle v,v\rangle}$	induced norm, length of a vector v
M, S	Manifold, submanifold
T_p	tangent plane at p
TM	Tangent bundle $TM = \cup_p T_p = \{(p,v), p\in M, v\in T_p\}$
$\mathfrak{F}(M)$	space of smooth functions on M
$\mathfrak{X}(M) = \Gamma(TM)$	space of smooth vector fields on M
vf	direction derivative of f with respect to vector v
$X,Y,Z \in \mathfrak{X}(M)$	Vector fields
$g \overset{\Sigma}{=} g_{ij}\mathrm{d}x_i \otimes \mathrm{d}x_j$	metric tensor (field)
$(\mathcal{U},x)$	local coordinates x in a chat $\mathcal{U}$
$\partial_i :=: \frac{\partial}{\partial x_i}$	natural basis vector
$\partial^i :=: \frac{\partial}{\partial x^i}$	natural reciprocal basis vector
∇	affine connection
$\nabla_X Y$	covariant derivative
$\prod_c^\nabla$	parallel transport of vectors along a smooth curve c
$\prod_c^\nabla v$	Parallel transport of $v \in T_{c(0)}$ along a smooth curve c
γ, γ_∇	geodesic, geodesic with respect to connection ∇
$\Gamma_{ij,l}$	Christoffel symbols of the first kind (functions)
Γ_{ij}^k	Christoffel symbols of the second kind (functions)
R	Riemann-Christoffel curvature tensor
$[X,Y]$	Lie bracket $[X,Y](f) = X(Y(f)) - Y(X(f)), \forall f \in \mathfrak{F}(M)$
∇-projection	$P_S = \arg\min_{Q\in S} D(\theta(P):\theta(Q))$
∇^*-projection	$P_S^* = \arg\min_{Q\in S} D(\theta(Q):\theta(P))$
C	Amari-Chentsov totally symmetric cubic 3-covariant tensor
$\mathcal{P} = \{p_\theta(x)\}_{\theta_i n \Theta}$	parametric family of probability distributions
$\mathcal{E}, \mathcal{M}, \Delta_D$	exponential family, mixture family, probability simplex
${}_{\mathcal{P}}I(\theta)$ Fisher information matrix	
${}_{\mathcal{P}}I(\theta)$	Fisher Information Matrix (FIM) for a parametric family $\mathcal{P}$